

- 1 (a) Fig 1.1 shows a fluid of density ρ in a rectangular container. The height of the liquid is h .

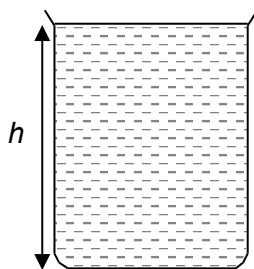


Fig 1.1

Show that the pressure P at the bottom of the container due to the fluid is given by

$$P = \rho gh$$

$$\begin{aligned} P &= \frac{F}{A} = \frac{W}{A} = \frac{mg}{A} \\ &= \frac{V\rho g}{A} \\ &= \frac{Ah\rho g}{A} \\ &= \rho gh \end{aligned}$$

[2]

- (b) (i) An object of mass m and density d is surrounded by air of density ρ .

Show that the resultant force F acting downward on the object is given by

$$F = mg \left(1 - \frac{\rho}{d} \right)$$

$$\begin{aligned} F_{\text{net}} &= F_{\text{weight}} - F_{\text{air}} \\ &= mg - F_{\text{air}} \\ &= mg - V\rho g \\ &= mg - m \left(\frac{\rho}{d} \right) \\ &= mg \left(1 - \frac{\rho}{d} \right) \end{aligned}$$

[2]

- (ii) A chemist uses an accurate balance to weigh a sample as shown in Fig. 1.2.

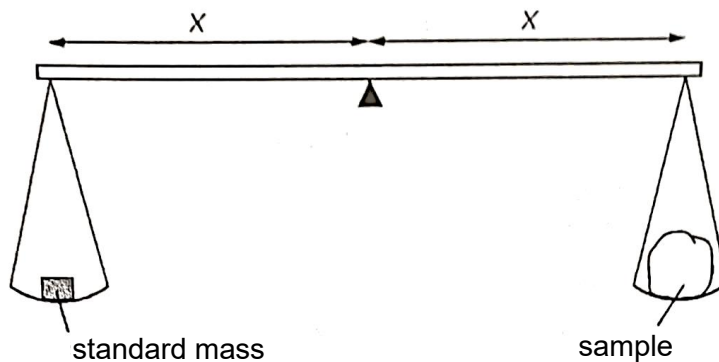


Fig. 1.2

The mass of the standard mass is 0.17851 kg. The density of the sample is 940.0 kg m^{-3} , the density of the standard mass is 8493 kg m^{-3} , and the density of air is 1.29 kg m^{-3} .

Using your answer in (b)(i), calculate the mass of the sample.

Leave your answer to 5 decimal places.

mass = 0.17851 kg [3]

[Total: 7 marks]

- 2 An object of mass m of 0.42 kg is attached to a spring S and the system is made to oscillate with simple harmonic motion on a horizontal, frictionless surface, as shown in Fig 2.1. The mass passes through the equilibrium position at P 200 times per minute.

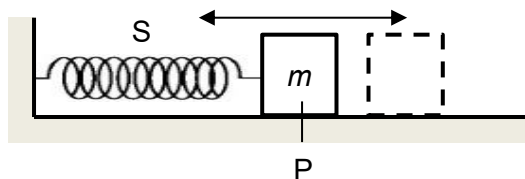


Fig 2.1

The kinetic energy of the mass as it passes through the equilibrium position is 500 mJ.

- (a) (i) Determine the period of the oscillation.

$$T = \frac{1}{f} = \frac{1}{200} = 0.005$$

period = 0.005 s [1]

- (ii) Show that the amplitude of the oscillation is approximately 15 cm.

[3]

- (iii) Sketch in Fig 2.2 the variation with time of the velocity of the mass for 2 cycles. Label the axes with appropriate values.



Fig 2.2

[2]

- (b) Deduce the change, if any, to the frequency of the oscillation if the following modifications are made separately to the experiment:

- (i) the experiment is done on Mars instead of the Earth,

.....
 [1]

- (ii) another spring identical to S is connected in parallel, as shown in Fig 2.3.

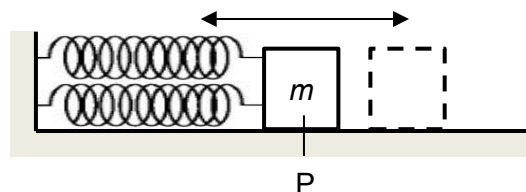


Fig 2.3

.....
 [1]

[Total: 8 marks]

- 3 (a) Define *gravitational potential* at a point.

.....
 [2]

- (b) The neutral point is the point where the gravitational field strength due to Earth is equal in magnitude and opposite in direction to the gravitational field strength due to the Moon.

The gravitational potentials at the Earth's and Moon's surfaces are $-62.3 \times 10^6 \text{ J kg}^{-1}$ and $-3.90 \times 10^6 \text{ J kg}^{-1}$ respectively.

A 10.0 kg mass projected from the Earth's surface needs $6.10 \times 10^8 \text{ J}$ of kinetic energy to just reach the neutral point from the Earth. The effects of air resistance may be neglected.

- (i) Determine the gravitational potential at the neutral point.

gravitational potential = J kg^{-1} [2]

- (ii) Determine the minimum kinetic energy needed to send a 1.4 kg rock from the surface of the Moon to the surface of the Earth.

minimum kinetic energy = J [3]

- (c) Two stars of mass M and $2M$, a distance of $3R$ apart, rotate in circles about their common centre of mass O .

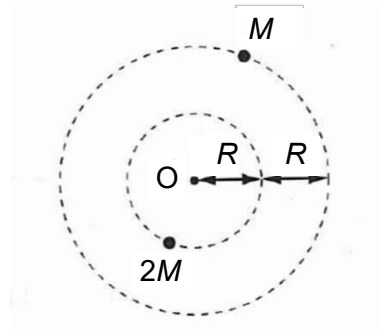


Fig 3.1

The period of this binary star system is 3.42×10^5 s. The value of M is 3.14×10^{30} kg.

- (i) Explain why the two stars experience the same magnitude of centripetal force.

.....

 [2]

- (ii) Determine the distance R .

$R = \dots\dots\dots$ m [3]

[Total: 12 marks]

- 4 (a) Waves from a point source pass through an area that is 1.6 cm wide, as shown in Fig. 4.1.

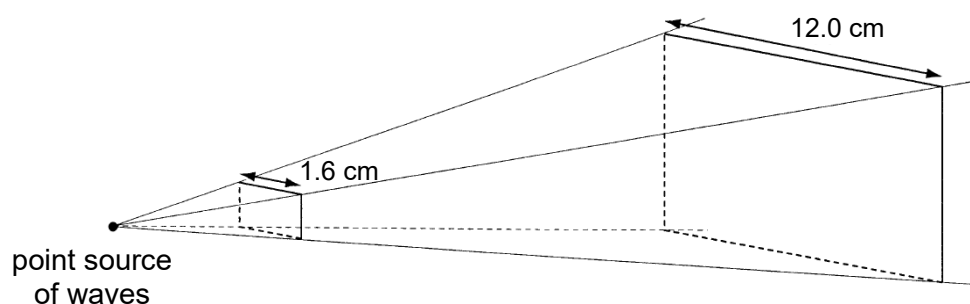


Fig. 4.1

Within this area, the intensity of the waves is I_0 and their amplitude is A_0 . The waves reach a second area of width 12.0 cm.

Determine the amplitude of the waves when they reach the second area in terms of A_0 .

Show your working clearly.

amplitude = [2]

- (b) A stationary wave is formed on a stretched string between two fixed points A and B.

The variation of the displacement y of particles of the string with distance x along the string for the wave at time $t = 0$ is shown on Fig. 4.2.

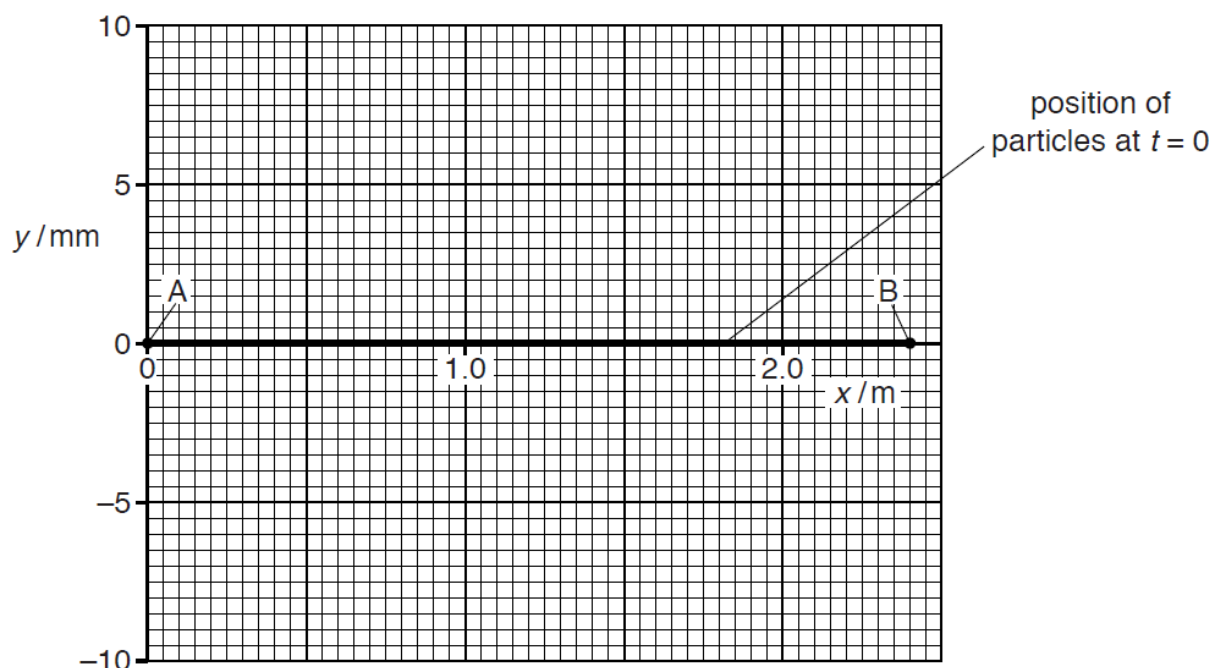


Fig. 4.2

The wave has a period of 20 ms and a wavelength of 1.2 m. The maximum amplitude of the particles of the string is 5.0 mm.

- (i) On Fig. 4.2, sketch the variation with position of the displacement of the string at
1. $t = 12.5$ ms (label this Y) [2]
 2. $t = 5.0$ ms (label this Z) [2]
- (ii) State the phase difference between the particles of the string at $x = 0.40$ m and at $x = 0.80$ m.

phase difference = [1]

[Total: 7 marks]

- 5 Two charged metal spheres A and B of diameters 18.0 cm and 12.0 cm respectively, are isolated in space, as shown in Fig. 5.1.

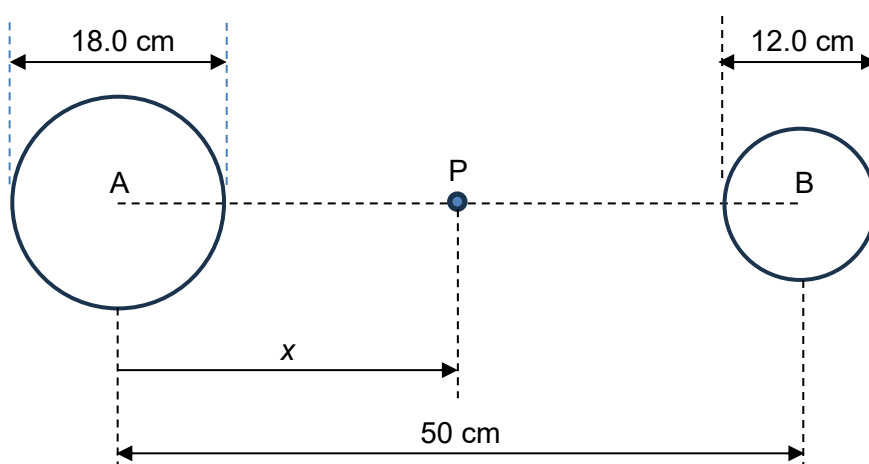


Fig. 5.1

The centres of the spheres are separated by a distance of 50.0 cm. Point P is at a distance x from the centre of sphere A along the line joining the centres of the two spheres.

The variation with x of the electric potential V at P is shown in Fig. 5.2.

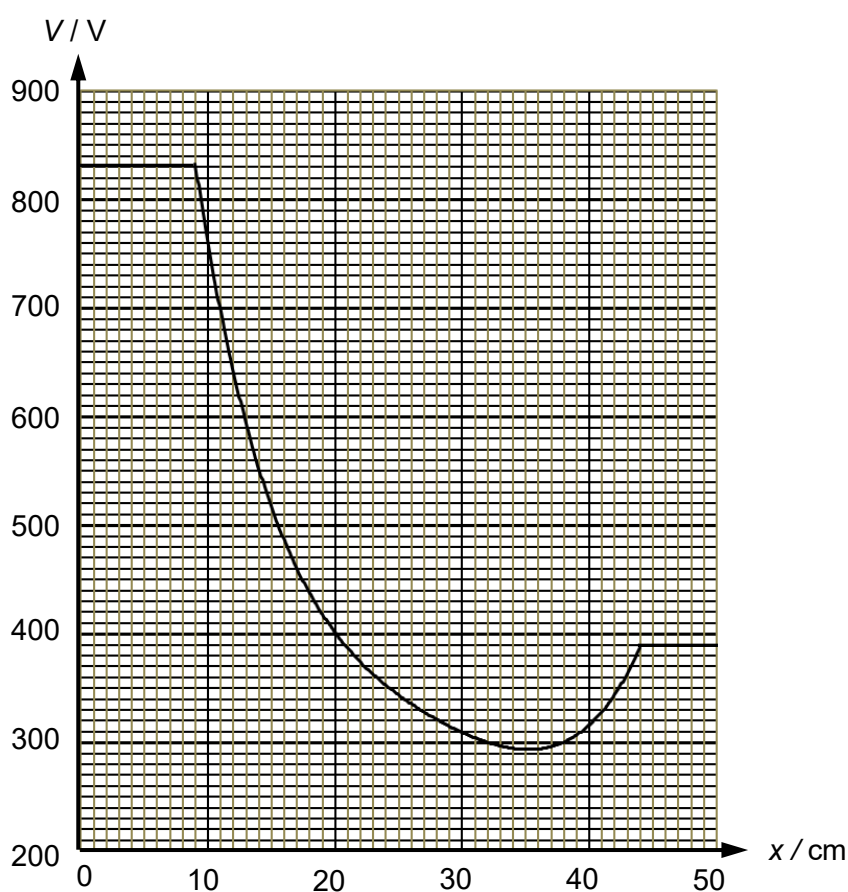


Fig. 5.2

- (a) (i) State and explain the direction of the electric field at the point P when $x = 25.0$ cm.

.....

 [2]

- (ii) Use Fig. 5.2 to determine the magnitude of the electric field strength on an electron placed at point P when $x = 25.0$ cm

electric field strength = V cm^{-1} [3]

- (iii) By making reference to electric fields, explain why the potential is constant between $x = 0.0$ cm and $x = 9.0$ cm.

.....

 [2]

- (b) A positively-charged ion is released from rest at $x = 25.0$ cm.

Describe the subsequent motion of the ion.

.....

 [2]

[Total: 9 marks]

- 6 In a metallic conductor, conduction electrons do not travel in a straight line through the conductor.

Fig. 6.1 shows some of the conduction electrons in a copper wire. The arrows represent the velocities of these electrons.

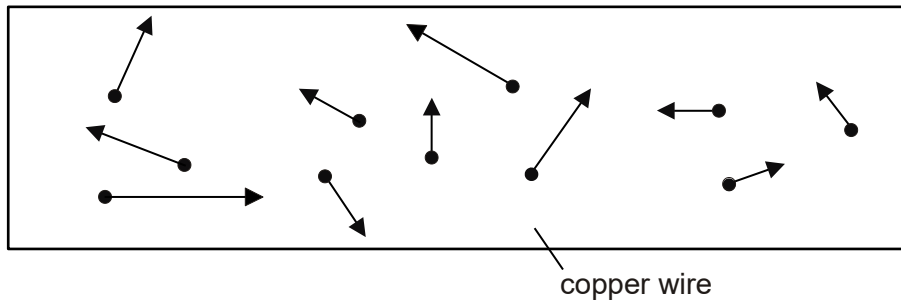


Fig. 6.1

- (a) Explain, by reference to the motion of the electrons, why there is no current in the wire.

.....
 [2]

- (b) An electric field is established inside the copper wire directed as shown in Fig. 6.2.

The dots represent electrons. The velocities of the electrons are not shown.

The average velocity that an electron travels along the conductor is called the drift velocity.

Draw on Fig. 6.2 an arrow to indicate the direction of the drift velocity of the electrons.

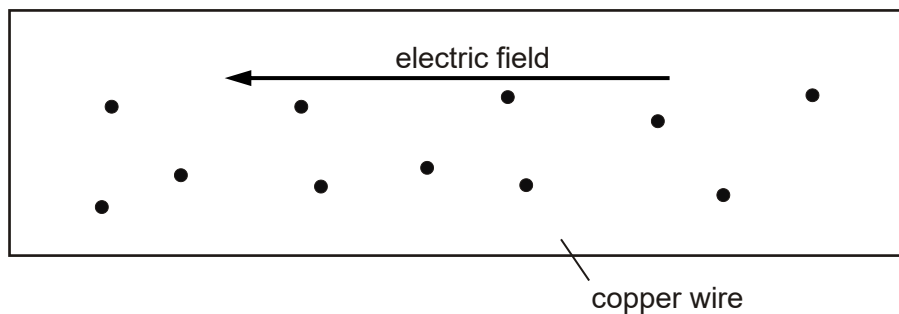


Fig. 6.2

[1]

- (c) In the circuit in Fig. 6.3, the length of the copper wire joining the negative terminal of the battery to the lamp is 0.50 m and has a radius of 0.40 mm. There are 8.5×10^{28} mobile electrons per cubic metre in copper.

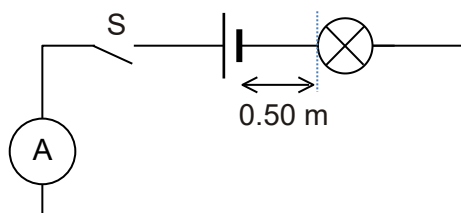


Fig. 6.3

- (i) When the switch S is closed, the ammeter reads 2.0 A. Calculate the average time it would take for an electron to move from the negative terminal of the battery to the lamp.

average time = s [2]

- (ii) The lamp lights up in a much shorter time than that calculated in (c)(i). Explain this observation.

.....

[1]

- (iii) The circuit is now connected with two copper wires of different thickness as shown in Fig. 6.4.

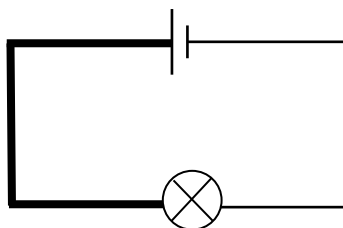


Fig 6.4

State and explain whether the drift velocity of electrons in the thicker wire is smaller than, equal to, or larger than that in the thinner wire.

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.....

..... [2]

[Total: 8 marks]

7 (a) State experimental evidence to suggest that the process of radioactive decay is

(i) random

.....
 [1]

(ii) spontaneous

.....
 [1]

(b) A student determines the half-life of the radioactive isotope of phosphorous-32. Phosphorous-32 decays by beta emission to form sulfur-32 which is stable.

The student measures the average count-rate from a sample of phosphorous-32 at various times t . The background-subtracted count rate, R , is shown in Fig. 7.1.

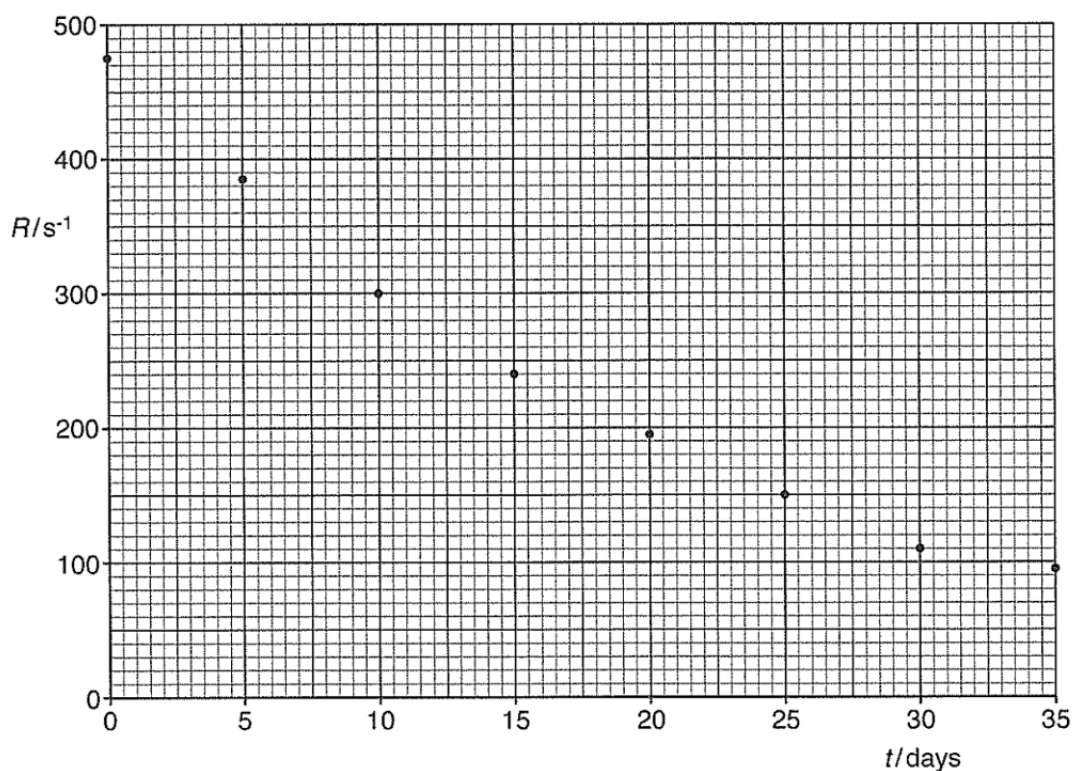


Fig. 7.1

(i) Use Fig. 7.1 to determine a value for the half-life of phosphorous-32. Show your working clearly.

half-life = days [3]

- (ii) Explain why, although the count rates are too low for the radiation to cause immediate symptoms in the student, careful shielding of the source is necessary.

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.....

.....

..... [2]

- (iii) Suggest why the determination of the half-life of phosphorus-32 by this method requires that the product of the decay is stable.

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.....

..... [1]

[Total: 8 marks]

8 Read the passage and answer the questions that follow.

Laser cooling of atoms

Atoms in a gas are always in motion. The temperature of a gas is related to the average kinetic energy of its atoms—the faster they move, the higher the temperature. Physicists have found ways to slow atoms down using laser light, and in doing so, cool the atoms to extremely low temperatures. This process is called laser cooling, illustrated in Fig. 8.1.

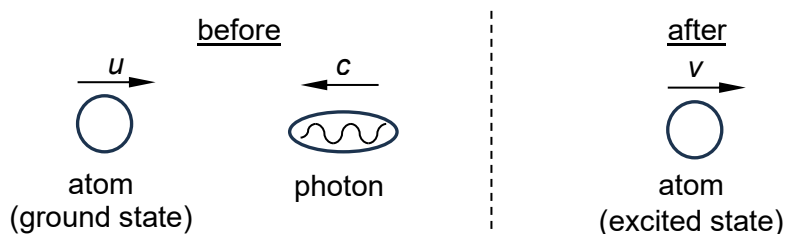


Fig. 8.1

Each photon has a small momentum. When the energy of the photon is just right, it can be absorbed by an atom, bringing it into an excited state. To slow an atom down, scientists shine light in the opposite direction that the atom is moving, so the atom absorbs photons, reducing its speed. Of course, after a short time, the atom will de-excite, emitting another photon – but this is in a random direction, rarely in the same direction as the first photon, and so generally the net effect is still that the atom's velocity in that axis is decreased.

Laser cooling experiments are often performed on rubidium (Rb) atoms. A simplified energy level diagram of the ^{87}Rb atom is shown in Fig. 8.2, showing the ground state, the first two excited states (labelled A and B), and their energies.

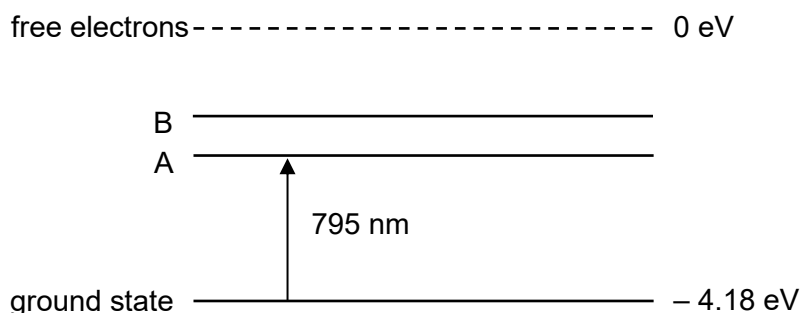


Fig. 8.2 (not to scale)

When an ^{87}Rb atom absorbs a photon of wavelength 795 nm, it transitions from the ground state to state A.

Table 8 shows some data about rubidium-87 (^{87}Rb).

atomic number	37
nucleon number	87
atomic mass	86.9 u
melting point at 1 atm	39.3 °C
boiling point at 1 atm	688 °C
lifetime of state A	27.6 ns
lifetime of state B	26.2 ns
emission wavelength from state A	795 nm
emission wavelength from state B	780 nm

Table 8

The setup of a laser cooling experiment is shown in Fig. 8.3 below.

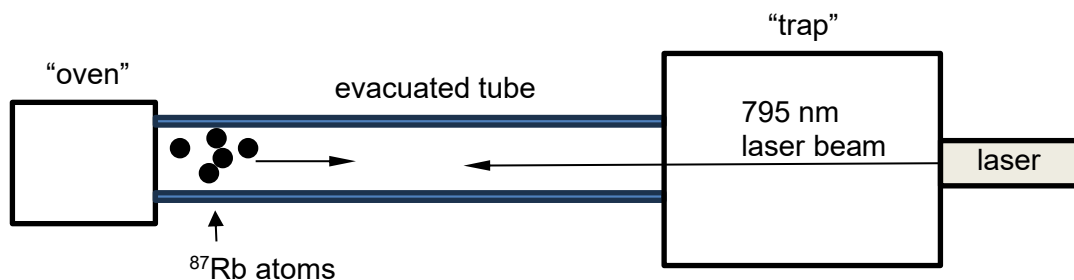


Fig. 8.3

A hot “oven” serves as a source of ^{87}Rb atoms, which travel through the evacuated tube towards the “trap”. A laser is directed in the opposite direction to the atoms’ motion, slowing them down significantly. When they reach the “trap”, they are slowed and cooled even further using other methods.

Laser cooling and other trapping methods have allowed physicists to make exciting discoveries, such as the first-ever creation of a Bose-Einstein Condensate in 1995 by Eric Cornell and Carl Wieman. A Bose-Einstein Condensate (BEC) is a new state of matter. In 1925, Bose and Einstein theorized that, as a consequence of wave-particle duality, at very low temperatures, the matter wave of atoms could have a wavelength λ greater than the average separation d between the atoms. When this happens, the matter waves of the individual atoms overlap to form a single wave, allowing the atoms to seemingly occupy the same space!

Of interest is the critical temperature T_c at which a BEC is formed, which depends on several factors, including the separation d between the atoms. This can be estimated using the relationship:

$$d \approx \frac{1}{\sqrt[3]{n}}$$

where n is the number of particles per unit volume.

Fig. 8.4 shows data from a laser cooling experiment when the temperature T of the cloud of atoms is $T > T_c$, $T \approx T_c$, and $T < T_c$.

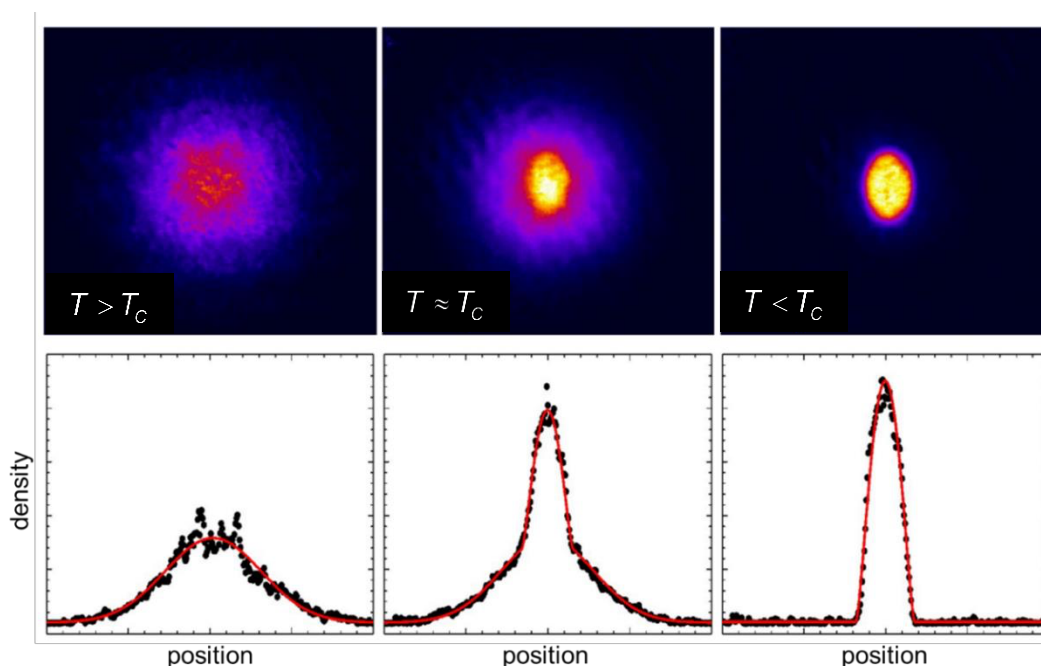


Fig. 8.4

- (a) (i) With reference to Fig. 8.2, explain why rubidium-87 atoms (^{87}Rb) only absorb photons of certain frequencies.

.....

[2]

- (ii) With reference to Fig. 8.2, determine the energy of state A in electron-volts.

energy = eV [3]

- (iii) When an ^{87}Rb atom transitions from state A to the ground state, a photon is released. Suggest why this photon is difficult to observe with the naked eye.

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[1]

- (b) A collection of ^{87}Rb atoms in the gaseous state emerge from the “oven” at a temperature of 1000 K.

Treating the ^{87}Rb atoms as an ideal gas, determine the average speed of an ^{87}Rb atom.

average speed = m s^{-1} [2]

- (c) Suggest what is meant by the “lifetime” of state A and state B in Table 8.

.....
[1]

- (d) (i) Using the principle of conservation of linear momentum, determine the magnitude of the change in momentum of an ^{87}Rb atom after it absorbs a photon from the laser, as in Fig. 8.1.

Show your working clearly.

change in momentum = kg m s^{-1} [3]

- (ii) Explain why, when an excited ^{87}Rb atom de-excites and emits a photon, the average change in momentum of the ^{87}Rb atom over many such emissions is zero.

.....
[1]

- (iii) Hence, using data from Table 8, determine the average force on the ^{87}Rb atom through the entire process of absorbing and emitting a photon.

force = N [2]

- (iv) Give one reason why the ^{87}Rb atoms could be cooled faster if the laser emitted photons that excite the atoms to state B instead of state A.

.....
[1]

- (e) State how Fig. 8.4 shows that a Bose-Einstein Condensate forms when the temperature T of the cloud of atoms is equal to or below the critical temperature T_c .

.....
[1]

- (f) The momentum of an ^{87}Rb atom in the “trap” is given by:

$$p = \sqrt{3mkT}$$

where m is the mass of an ^{87}Rb atom, T is the thermodynamic temperature of the collection of atoms, and k is the Boltzmann constant.

Using the de Broglie relation, calculate the critical temperature T_c at which a collection of ^{87}Rb atoms forms a BEC in a “trap” where $n = 1.00 \times 10^{19} \text{ m}^{-3}$.

$T_c = \dots\dots\dots \text{ K} \quad [4]$

[Total: 21 marks]

End of paper

Answer **all** questions in the spaces provided.

- 1 A car of mass 1700 kg travels over a curved hump in the road as shown in Fig. 1.1. The radius of curvature of the hump is 45 m.

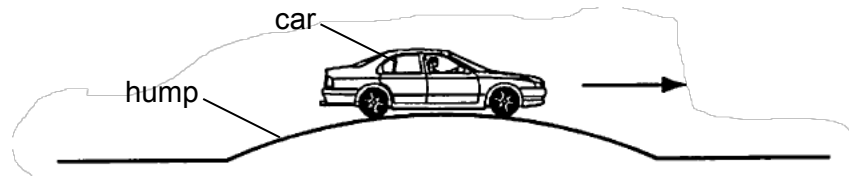


Fig. 1.1

- (a) The speed of the car at the top of the hump is 19 m s^{-1} .

Determine, for the car at the top of the hump,

- (i) the magnitude of the centripetal force acting on the car,

centripetal force = N [1]

- (ii) the magnitude of the normal contact force exerted by the road on the car.

normal contact force = N [2]

- (b) Determine the maximum speed v_{max} that the car can travel at without losing contact with the top of the hump. Explain your working.

$$v_{max} = \dots\dots\dots \text{ m s}^{-1} \quad [3]$$

[Total: 6]

- 2 A long, straight wire W carrying a direct current of 3.0 A flows in the direction as shown in Fig. 2.1.

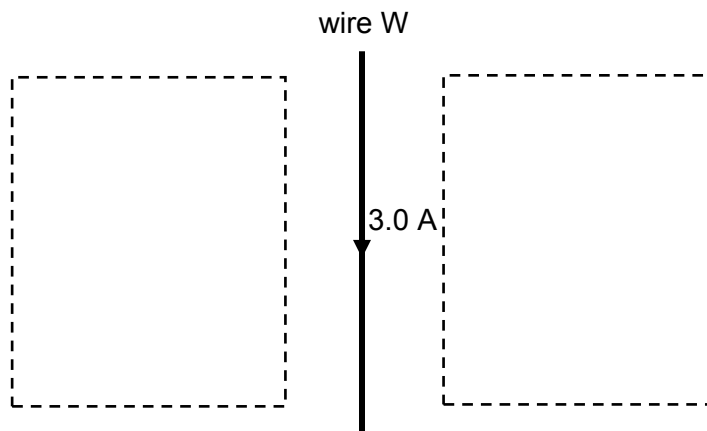


Fig. 2.1

- (a) Draw on Fig. 2.1, the pattern of the magnetic field produced by wire W in the regions indicated by the dotted boxes. Use the symbol $\times$ to represent magnetic field directed into the page and use the symbol $\bullet$ to represent magnetic field directed out of the page. [3]
- (b) A similar wire Y is placed parallel to wire W, separated by a distance of 40.0 cm as shown in Fig. 2.2. Initially, there is no current in wire Y.

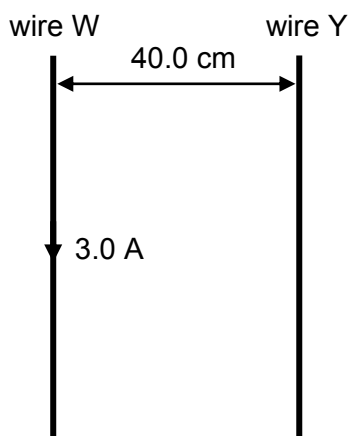


Fig. 2.2

- (i) Show that the magnetic flux density at wire Y due to the current in wire W is $1.5 \times 10^{-6} \text{ T}$.

- (ii) A current of 1.0 A is now switched on in wire Y and flows in the opposite direction as the direction of current flow in wire W.

Use your answer in (b)(i) to calculate the force per unit length acting on wire Y.

force per unit length = N m⁻¹ [2]

- (iii) Explain why the force that the two wires exert on each other is repulsive.

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..... [3]

- (iv) Determine a possible position, other than at infinity, where the resultant magnetic flux density due to the magnetic fields of both wires is zero.

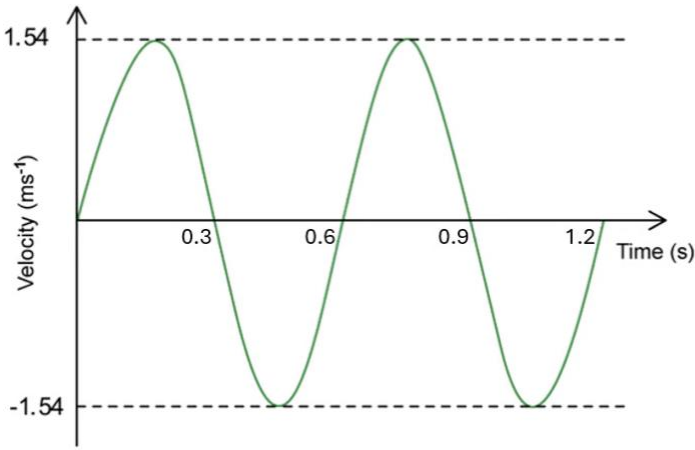
position: [3]

[Total: 12]

[Turn over

HCI H2 Physics 9749
2025 C2 Preliminary Examination
Paper 2 Suggested Solutions

Q1	
(a)	Pressure p at depth h in the fluid = $\frac{\text{force } F}{\text{area of base of container } A}$ = $\frac{\text{weight of fluid column } W}{\text{area } A}$ = $\frac{\text{mass of fluid column } m \times \text{acceleration of free fall } g}{A}$ = $\frac{(\text{density } \rho \times \text{volume } V)g}{A}$ = $\frac{\rho Ahg}{A}$ = ρgh
(b)(i)	Let the volume of the object be V. Resultant force F = weight of object mg – upthrust U $F = mg - U$ = $(Vd)g - V\rho g$ = $Vdg\left(1 - \frac{\rho}{d}\right)$ = $mg\left(1 - \frac{\rho}{d}\right)$
(b)(ii)	At equilibrium, taking moments about the pivot, Clockwise moment due to sample = anticlockwise moment due to standard mass (Resultant force on sample) $\times$ = (Resultant force on standard mass) $\times$ $mg\left(1 - \frac{\rho}{d}\right) = m_s g\left(1 - \frac{\rho}{d_s}\right)$ $m = m_s\left(1 - \frac{\rho}{d_s}\right) / \left(1 - \frac{\rho}{d}\right)$ = $(0.17851)\left(1 - \frac{1.29}{8493}\right) / \left(1 - \frac{1.29}{940.0}\right)$ = 0.17873 kg (5 d.p.)

Q2	
(a)(i)	The mass passes through the equilibrium position P twice in a cycle, meaning that there are 100 complete cycles per minute. period, $T = 1/f = 60/100 = 0.600$ s
(a)(ii)	At equilibrium position $x = 0$, the kinetic energy of the mass E_k is at its maximum. $\omega = \frac{2\pi}{T} = \frac{2\pi}{0.600} = 10.47 \text{ rad s}^{-1}$ $E_k = \frac{1}{2}m\omega^2 x_o^2 = \frac{1}{2}(0.42)(10.47)^2 x_o^2$ $\text{Amplitude } x_o = \sqrt{\frac{2(0.500)}{0.42(10.47)^2}} = 0.147 \text{ m} = 14.7 \text{ cm} = 15 \text{ cm (2 s.f.)}$
(a)(iii)	$E_k = \frac{1}{2}mv^2 \Rightarrow v = \sqrt{\frac{2E_k}{m}} = \sqrt{\frac{2(0.500)}{0.42}} = 1.54 \text{ m s}^{-1}$  • Sinusoidal waveform • Maximum velocity of 1.54 m s^{-1} and time period of 0.60 s <i>*No information is given whether mass starts oscillating from its equilibrium position or at its amplitude – sine or cosine graph</i>
(b)(i)	No change
(b)(ii)	Frequency multiplied by $\sqrt{2}$.

Q3	
(a)	Gravitational potential at a point is the work done per unit mass in bringing a small test mass from infinity to that point (without a change in kinetic energy)
(b)(i)	To just reach the neutral point from the Earth, gain in gravitational potential energy GPE = Loss in kinetic energy $\Delta U = m\Delta\phi = -\Delta KE$ $10.0\Delta\phi = 6.10 \times 10^8 \text{ J}$ $\Delta\phi = 6.10 \times 10^7 \text{ J kg}^{-1}$ $\phi_{\text{neutral point}} - \phi_{\text{earth surface}} = 6.10 \times 10^7$ $\phi_{\text{neutral point}} = 6.10 \times 10^7 + \phi_{\text{earth surface}}$ $\phi_{\text{neutral point}} = 6.10 \times 10^7 + (-62.3 \times 10^6)$ $\phi_{\text{neutral point}} = -1.3 \times 10^6 \text{ J kg}^{-1} \text{ (2 s.f.)}$ OR total energy at Earth surface = total energy at neutral point $KE_{\text{earth surface}} + U_{\text{earth surface}} = KE_{\text{neutral point}} + U_{\text{neutral point}}$ $6.10 \times 10^8 + 10(-62.3 \times 10^6) = 0 - 10.0 \times \phi_{\text{neutral point}}$ $\phi_{\text{neutral point}} = -1.30 \times 10^6 \text{ J kg}^{-1} \text{ (3 s.f.)}$
(b)(ii)	The rock from the Moon must have enough energy to go past the neutral point, then resultant gravitational force of the Earth-mass system on the mass will accelerate it to the Earth. gravitational potential difference between the Moon's surface and the neutral point $\Delta\phi = \phi_{\text{neutral point}} - \phi_{\text{moon surface}}$ $= -1.30 \times 10^6 - (-3.90 \times 10^6)$ $= 2.60 \times 10^6 \text{ J kg}^{-1}$ Hence, the minimum kinetic energy needed to send a 1.4 kg rock from the Moon to the neutral point is $m\Delta\phi = (1.4)(2.60 \times 10^6) = 3.64 \times 10^6 \text{ J}$ or $3.6 \times 10^6 \text{ J}$ (2 s.f.)
(c)(i)	The gravitational force exerted on one star by the other star provides the centripetal force for each orbit. This pair of forces is an action-reaction pair (Newton's 3rd law), always equal in magnitude (and opposite in direction). OR The gravitational force between stars provides the centripetal force for each orbit. By Newton's Law of Gravitation, the gravitation force between the stars = $\frac{GM(2M)}{(3R)^2}$, thus the two stars experience the same magnitude of centripetal force.

(c)(ii)

$$\omega = \frac{2\pi}{T} = \frac{2\pi}{3.42 \times 10^5} = 1.837 \times 10^{-5} \text{ rad s}^{-1}$$

Consider the star of mass M ,

The gravitational force due to star of mass $2M$ provides the centripetal force for the orbit of M .

$$F_G = Ma_c$$

$$\frac{GM(2M)}{(3R)^2} = M(2R)\omega^2$$

$$R = \sqrt[3]{\frac{GM}{9\omega^2}}$$

$$= \sqrt[3]{\frac{6.67 \times 10^{-11} (3.14 \times 10^{30})}{9(1.837 \times 10^{-5})^2}}$$

$$= 4.10 \times 10^9 \text{ m}$$

Q4	
(a)	Let the distance from the source to the first area be r. Using geometry, the distance from the point source is 7.5 times more for the second area. Method I $I \propto \frac{1}{r^2} \propto A^2$. Hence, amplitude is inversely proportional to the distance away from a point source, $A \propto \frac{1}{r} \quad [\text{M1}]$ Since the distance from the point source is 7.5 times more for the second area, the amplitude would be 7.5 times less. Hence, the new amplitude would be $A_2 = \frac{A_0}{7.5} = 0.13A_0 \quad [\text{A1}]$ Method II Since the source is a point source, the intensity of the wave is inversely proportional to the square of the distance that the waves travel, $I \propto \frac{1}{r^2}$. Hence the intensity at the second area, I_2 $\frac{I}{I_2} \propto \frac{r_2^2}{r^2} = \frac{(7.5r)^2}{(r)^2}$ $I_2 = \frac{I}{7.5^2} \quad [\text{M1}]$ Intensity is directly proportional to the square of the amplitude. Hence, the amplitude: $\frac{I}{I_2} \propto \frac{A_0^2}{A_2^2} \rightarrow A_2^2 = \frac{I_2}{I} A_0^2$ $A_2 = \frac{A_0}{7.5} = 0.13A_0 \quad [\text{A1}]$ Method III Intensity = Power / Area Intensity $\propto$ (1 / Area) $\propto$ (amplitude)² $\frac{I'}{I} = \frac{S}{S'} = \left(\frac{1.6}{12}\right)^2 \rightarrow I' = \left(\frac{1.6}{12}\right)^2 I \quad [\text{M1}]$ $A' = \frac{1.6}{12} A_0 = 0.13A_0 \quad [\text{A1}]$

(b)(i)
1 & 2

Wavelength = 1.2 m. **Two** full wavelengths of **stationary waves** must be drawn with

- Nodes at 0.0 m, 0.6 m, 1.2 m, 1.8 m and 2.4 m.
- Antinode peaks at 0.3 m and 1.5 m or at 0.9 m and 2.1 m [B1]

Maximum amplitude for Y at 12.5 ms

$$y = 5.0 \sin(\omega t) = 5.0 \sin\left(\frac{2\pi}{T} t\right)$$

[B1]

$$y = 5.0 \sin\left(\frac{2\pi}{20} 12.5\right) = -3.5 \text{ mm}$$

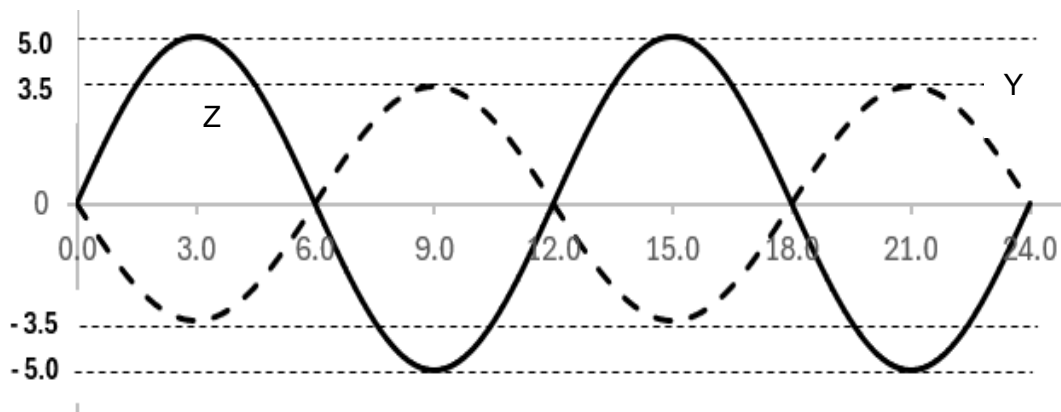
Maximum amplitude for Z at 5.0 ms

$$y_z = 5.0 \sin(\omega t) = 5.0 \sin\left(\frac{2\pi}{20} 5.0\right)$$

[B1]

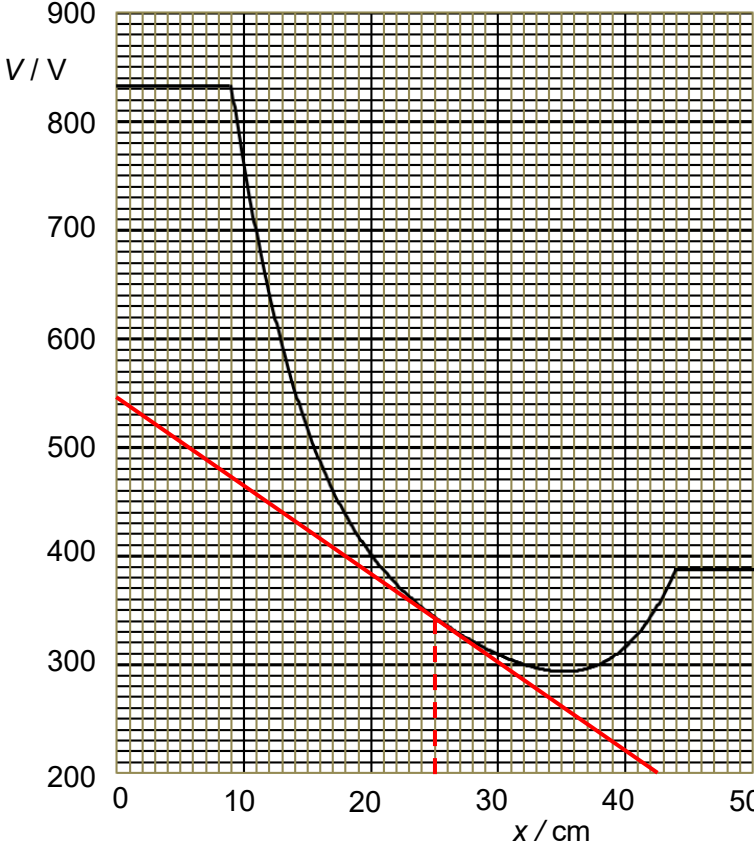
$$y_z = 5.0 \text{ mm}$$

The antinode peaks for Z should coincide with antinode troughs for Y and vice versa [B1]



(b)(ii)

180° or π rad

Q5	
(a)(i)	Direction of the electric field at $x = 25.0$ cm is towards the right. Electric field is always directed towards lower potential. OR Direction of the electric field at $x = 25.0$ cm is towards the right. The potential gradient is negative at this value of x as can be observed by drawing a tangent to the curve at $x = 25.0$ cm. Since E is (the negative) of the potential gradient, E takes on a positive value, which means the electric field is directed towards the right.
(a)(ii)	 Draw a tangent to the curve at $x = 25.0$ cm Potential gradient $\frac{dV}{dx} = \frac{545 - 200}{0.0 - 42.5} = -8.12$ (3 s.f.) Electric field $E = -\frac{dV}{dx} = 8.12 \text{ V cm}^{-1}$ (acceptable range: 7.3 – 9.7)
(a)(iii)	Mobile charge carriers within a conductor will always re-distribute until a certain equilibrium state where the electric field within it is zero. Zero electric field means that there is zero potential gradient, and hence constant potential.
(b)	The ion will accelerate to the right until $x = 35.0$ cm, then the ion will decelerate and momentarily come to rest at $x = 42.0$ cm, and accelerate back to the left and momentarily stops at $x = 25.0$ cm before accelerating to the right again, repeating the motion.

	OR	
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	The ion will oscillate between the points $x = 25.0 \text{ cm}$ and $x = 42.0 \text{ cm}$.	
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Q6	
(a)	The motion of electrons is random, so there is no net flow of electrons in any direction. OR On average, as much mobile electrons move in one direction as in the opposite direction, Thus there is no net transfer / movement of electric charge in a particular direction.
(b)	Arrow to the right.
(c)(i)	$v = \frac{I}{neA} = \frac{2.0}{8.5 \times 10^{28} (1.60 \times 10^{-19}) \pi (0.40 \times 10^{-3})^2} = 2.9 \times 10^{-4} \text{ m s}^{-1}$ $t = L/v = 0.50 / (2.9 \times 10^{-4}) = 1700 \text{ s (or 1710 s to 3 s.f.)}$ <i>Accept answer up to 3 significant figures.</i>
(c)(ii)	All mobile electrons in the circuit start drifting at the same time as the electric field is established in the wire almost instantaneously. The lamp lights up as soon as the mobile electrons <u>already</u> in the lamp filament begin to move which is much shorter than the time calculated in (c)(i).
(c)(iii)	As $I = nve$ and current (I), charge density (n) and charge (e) are the same in both wires, drift velocity v is inversely proportional to the cross-sectional area A of the wires. As the cross-sectional area for thicker wire is larger than that of the thinner wire, drift velocity of the electrons in the thicker wire is smaller than that in the thinner wire. (no marks for wrong explanations) Alternatively, Since the current in the circuit is the same, the thicker wire having a larger cross-sectional area (A), has greater number of electrons per unit length, since the charge density (n) is the same for the same metal. Thus the thicker wire should have a smaller drift velocity (v) to keep the number of electrons passing through a point per unit time (i.e. the current I) constant.

Q7	
(a)(i)	When a graph showing the variation of count-rate from a radioactive sample over time is plotted, we observe fluctuations instead of a smooth curve.
(a)(ii)	Variations in external factors such as temperature and pressure do not affect the curve variation in the count-rate over time
(b)(i)	From the figure, at $t = 0$, $C_0 = 475 \text{ s}^{-1}$. When the count rate is halved, $C_t = \frac{475}{2} = 237.5 \text{ s}^{-1}$. To reach this value, reading from the graph, half-life is 15 days At $t = 10$ days, $C = 300 \text{ s}^{-1}$. This is halved at $C = 150 \text{ s}^{-1}$, which is at $t = 25$ days. This also gives a half-life of 15 days. The (averaged) half life is thus 15 days.
(b)(ii)	Beta particles have a sufficiently long range that could cross the entire lab, so the student is exposed to it even when not conducting the experiment. OR Beta particles are ionising radiation that can cause health problems like cancer when exposed to a sufficiently high dose. Radiation dosage looks at the total amount accumulated over time and if this is high enough due to prolonged exposure, it can be a health hazard.
(b)(iii)	If the product is not stable, it can decay and that will contribute to the count rate measured which will not be just due to the decay of phosphorous-32 alone.

Q8	
(a)(i)	The energy E of a photon is proportional to its frequency ($E = hf$). The atom can only absorb a photon of energy equal to <u>the difference in two energy levels</u>. From Fig 8.2, we see that <u>the rubidium atom has discrete energy levels</u>, and thus has discrete energy differences, thus only certain frequencies of photons can be absorbed.
(a)(ii)	Energy of photon, $\Delta E = hf = \frac{hc}{\lambda} = \frac{(6.63 \times 10^{-34})(3.00 \times 10^8)}{795 \times 10^{-9}} = 2.502 \times 10^{-19} \text{ J } (= 1.56 \text{ eV})$ Energy difference, $E_A - E_{\text{ground}} = \Delta E$ $E_A - (-4.18)(1.60 \times 10^{-19}) = 2.502 \times 10^{-19}$ $E_A = -4.186 \times 10^{-19} \text{ J}$ $= -2.62 \text{ eV}$
(a)(iii)	The photon is in the <u>infrared</u> part of the electromagnetic spectrum, outside the visible range, so it cannot be seen with the naked eye. OR The visible spectrum is from <u>400 nm to 700 nm / 750 nm</u>, thus the photon's wavelength lies outside the visible range. OR The visible spectrum has <u>wavelengths on the order of 10^{-7} m</u>, and the photon's wavelength is <u>near the edge of this range</u>. OR (not intended solution but accepted due to phrasing of the question) A <i>single</i> photon reaching the eye has a very small intensity, thus it will be hard to see/detect with the naked eye.
(b)	$E = \frac{3}{2} kT = \frac{1}{2} mv^2$ $v = \sqrt{\frac{3kT}{m}} = \sqrt{\frac{3(1.38 \times 10^{-23})(1000)}{86.9 \times 1.66 \times 10^{-27}}}$ $= 535.72 \approx 536 \text{ m s}^{-1}$ OR $PV = NkT = \frac{1}{3} Nm \langle c^2 \rangle$ $\sqrt{\langle c^2 \rangle} = \sqrt{\frac{3kT}{m}} = \sqrt{\frac{3(1.38 \times 10^{-23})(1000)}{86.9 \times 1.66 \times 10^{-27}}}$ $= 535.72 \approx 536 \text{ m s}^{-1}$

(c)	It is the (average) time that the atom stays in that state before it de-excites to another state.	
(d)(i)	Using the de Broglie relation, $p_{\text{photon}} = \frac{h}{\lambda} = \frac{6.63 \times 10^{-34}}{795 \times 10^{-9}} = 8.34 \times 10^{-28} \text{ kg m s}^{-1}$ OR $E = hf = \frac{hc}{\lambda} = pc$ From (a)(ii), $E = 2.502 \times 10^{-19} \text{ J}$ $\therefore p_{\text{photon}} = \frac{E}{c} = \frac{2.502 \times 10^{-19}}{3.00 \times 10^8} = 8.34 \times 10^{-28} \text{ kg m s}^{-1}$ By the principle of conservation of momentum, taking right as positive $p_{\text{atom}} - p_{\text{photon}} = p'_{\text{atom}}$ $\Delta p_{\text{atom}} = p'_{\text{atom}} - p_{\text{atom}} = -p_{\text{photon}}$ Thus $\Delta p_{\text{atom}} = \left -(8.34 \times 10^{-28}) \right$ $= 8.34 \times 10^{-28} \text{ kg m s}^{-1}$	
(d)(ii)	The photons are emitted in randomly in all directions. <u>Since momentum is a vector quantity</u>, the average change in momentum from this process is the <u>vector sum</u> $\Delta \vec{p} = \frac{\Delta \vec{p}_1 + \Delta \vec{p}_2 + \dots + \Delta \vec{p}_N}{N} = \frac{0}{N} = 0$	
(d)(iii)	From Table 8, the lifetime of the atom in state A is 27.6 ns. Since from (d)(ii) we know that, on average, only absorbing a photon causes a change in momentum, therefore, $F = \frac{\Delta p}{\Delta t}$ $= \frac{8.34 \times 10^{-28}}{27.6 \times 10^{-9}}$ $= 3.02 \times 10^{-20} \text{ N}$	
(d)(iv)	The photons required to excite the atom to state B have shorter wavelength and hence greater momentum, thus <u>each interaction causes a greater decrease in momentum of the atom</u>. OR The lifetime of state B is shorter than state A, so <u>it can absorb more photons per unit time</u> OR <u>the average force on the atom is greater.</u>	
(e)	From Fig. 8.4, when the temperature decreases to T_c, <u>the density of the cloud of atoms increases significantly</u>, suggesting that the particles are overlapping. OR	

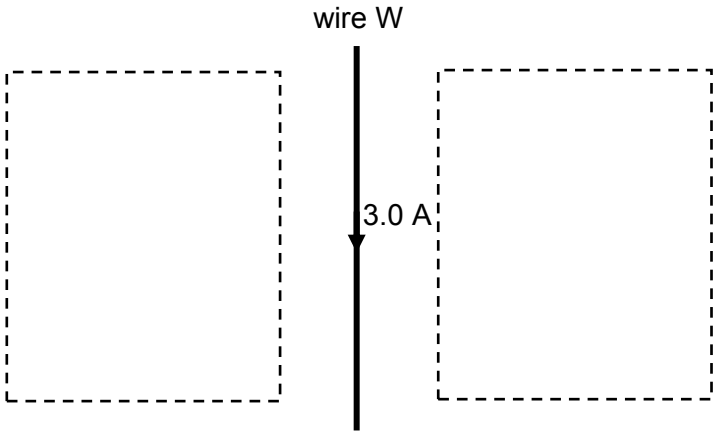
	From Fig. 8.4, when the temperature is equal to or below T_c , <u>almost all the particles occupy a small space in the middle of the trap, suggesting that the particles are overlapping.</u>	
(f)	From the de Broglie relation, $\lambda = \frac{h}{p} = \frac{h}{\sqrt{3mkT}}$ Since a BEC forms when $\lambda \geq d$, where $T \leq T_c$, and $d \approx \frac{1}{\sqrt[3]{1.00 \times 10^{19}}} = 4.6416 \times 10^{-7} \text{ m}$, $\frac{h}{\sqrt{3mkT_c}} = \frac{1}{\sqrt[3]{n}}$ $T_c = \frac{h^2}{3mk} n^{\frac{2}{3}}$ $= \frac{(6.63 \times 10^{-34})^2}{3(86.9 \times 1.66 \times 10^{-27})(1.38 \times 10^{-23})} (1.00 \times 10^{19})^{\frac{2}{3}}$ $= 3.4164 \times 10^{-7} \approx 3.42 \times 10^{-7} \text{ K}$ <u>Alternative solution without using de Broglie relation (max 3 marks only)</u> From the Heisenberg uncertainty principle, $\Delta x \Delta p \gtrsim h$ Taking $\Delta p \approx p = \sqrt{3mkT}$, $\Delta x \gtrsim \frac{h}{\sqrt{3mkT}}$ Since a BEC forms when the particles “overlap”, i.e. $\Delta x \geq d$, where $T \leq T_c$, and $d \approx \frac{1}{\sqrt[3]{1.00 \times 10^{19}}} = 4.6416 \times 10^{-7} \text{ m}$, $\frac{1}{\sqrt[3]{n}} = \frac{h}{\sqrt{3mkT_c}}$ $T_c = \frac{h^2}{3mk} n^{\frac{2}{3}}$ $= \frac{(6.63 \times 10^{-34})^2}{3(86.9 \times 1.66 \times 10^{-27})(1.38 \times 10^{-23})} (1.00 \times 10^{19})^{\frac{2}{3}}$ $= 3.4164 \times 10^{-7} \approx 3.42 \times 10^{-7} \text{ K}$	

Answer **all** questions in the spaces provided.

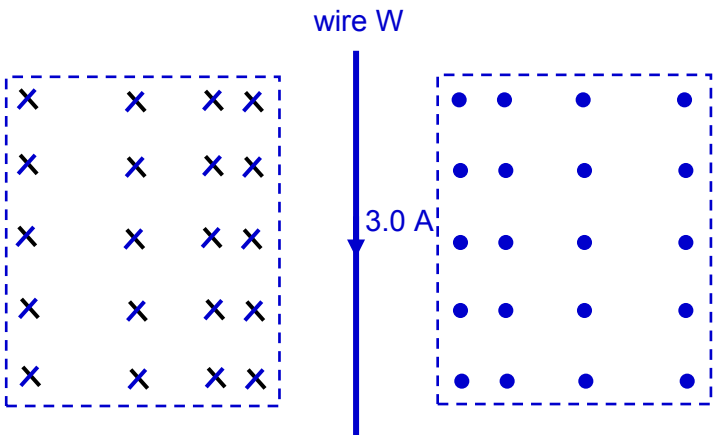
1	A car of mass 1700 kg travels over a curved hump in the road as shown in Fig. 1.1. The radius of curvature of the hump is 45 m. Fig. 1.1
	(a) The speed of the car at the top of the hump is 19 m s⁻¹. Determine, for the car at the top of the hump,
	(i) the magnitude of the centripetal force acting on the car,
	centripetal force = N [1]
L1	$F_c = \frac{mv^2}{r} = \frac{(1700)(19)^2}{45} = 13638 = \mathbf{14000\text{ N}}$ A1
	(ii) the magnitude of the normal contact force exerted by the road on the car.
	normal contact force = N [2]
L2	Free body diagram: The resultant force of the downward weight of the car W and the upward normal contact force exerted by the road N provide for the centripetal force required. $W - N = 13638$ $(1700)(9.81) - N = 13638$ $N = 3039 = \mathbf{3000\text{ N (magnitude)}}$ Direction of N: vertically upwards M1 A1
	(b) Determine the maximum speed v_{max} that the car can travel at without losing contact with the top of the hump. Explain your working.
	$v_{max} = \text{ m s}^{-1}$ [3]
L2	The resultant force of the downward weight of the car W and the upward normal contact force exerted by the road N provide for the centripetal force F_c required. Thus, $W - N = F_c$ $N = W - F_c$ B1

	For the car not to lose contact with the top of the hump, the normal contact force N with the road must be greater than zero, i.e. $N > 0$ $W - F_c > 0$ $mg - \frac{mv^2}{r} > 0$ $v < \sqrt{rg}$ $v < \sqrt{(45)(9.81)}$ $v_{max} = 21.011 = 21 \text{ m s}^{-1}$	B1 A1
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[Total: 6]

2	A long, straight wire W carrying a direct current of 3.0 A flows in the direction as shown in Fig. 2.1.  Fig. 2.1	
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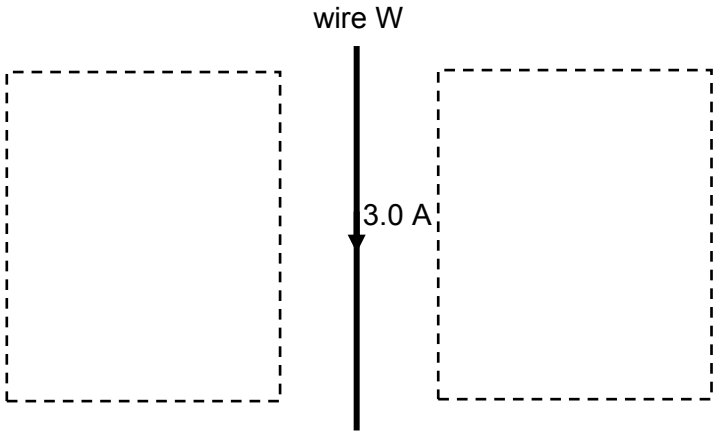
(a)	Draw on Fig. 2.1, the pattern of the magnetic field produced by wire W in the regions indicated by the dotted boxes. Use the symbol $\times$ to represent magnetic field directed into the page and use the symbol $\bullet$ to represent magnetic field directed out of the page.	[3]
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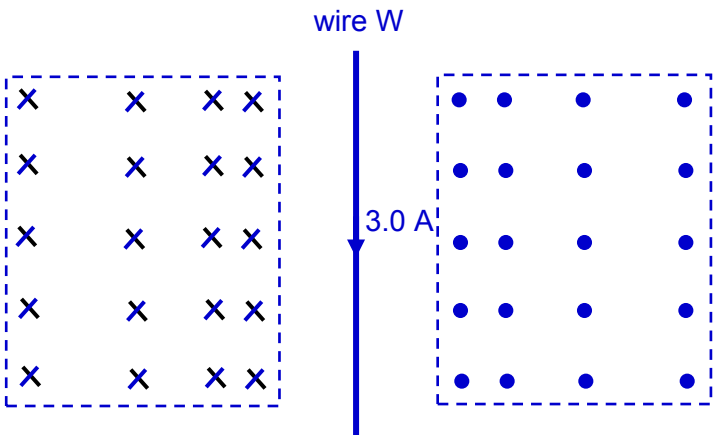
L1	Solution:  Marks scheme: 1 mark for: - Correct direction of magnetic flux density on left side (into plane of paper) and right side (out of plane of paper) of wire W. 1 mark for: - Increasing spacing with increasing distance from wire W. 1 mark for (Pattern shown sufficiently) - At least 4 columns each of $\times$ and $\bullet$ shown to show the trend of increasing spacing between consecutive circular field lines.	B1 B1 B1
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[Turn over]

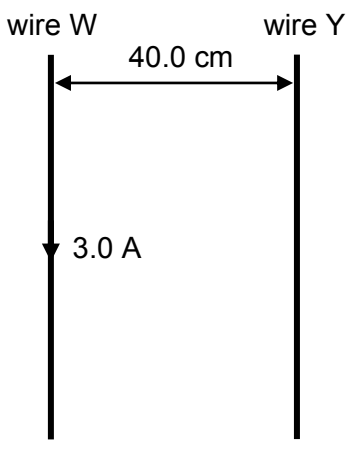
	For the car not to lose contact with the top of the hump, the normal contact force N with the road must be greater than zero, i.e. $N > 0$ $W - F_c > 0$ $mg - \frac{mv^2}{r} > 0$ $v < \sqrt{rg}$ $v < \sqrt{(45)(9.81)}$ $v_{max} = 21.011 = 21 \text{ m s}^{-1}$	B1 A1
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[Total: 6]

2	A long, straight wire W carrying a direct current of 3.0 A flows in the direction as shown in Fig. 2.1.  Fig. 2.1	
	(a) Draw on Fig. 2.1, the pattern of the magnetic field produced by wire W in the regions indicated by the dotted boxes. Use the symbol x to represent magnetic field directed into the page and use the symbol • to represent magnetic field directed out of the page.	[3]

L1	Solution:  Marks scheme: 1 mark for: - Correct direction of magnetic flux density on left side (into plane of paper) and right side (out of plane of paper) of wire W. 1 mark for: - Increasing spacing with increasing distance from wire W. 1 mark for (Pattern shown sufficiently) - At least 4 columns each of x and • shown to show the trend of increasing spacing between consecutive circular field lines.	B1 B1 B1
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[Turn over]

		At least 3 rows each of x and • shown to show the trend of equal spacing parallel to the wire.	
	(b)	A similar wire Y is placed parallel to wire W, separated by a distance of 40.0 cm as shown in Fig. 2.2. Initially, there is no current in wire Y.  Fig. 2.2	
	(i)	Show that the magnetic flux density at wire Y due to the current in wire W is 1.5×10^{-6} T.	
			[1]
	L1	Magnetic flux density $B = \frac{\mu_0 I}{2\pi d}$ where I is the current in wire W and d the distance of Y from W. $B = \frac{(4\pi \times 10^{-7})(3.0)}{2\pi(40.0 \times 10^{-2})}$ $= 1.5 \times 10^{-6} \text{ T (Shown)}$	M1 A0
	(ii)	A current of 1.0 A is now switched on in wire Y and flows in the opposite direction as the direction of current flow in wire W. Use your answer in (b)(i) to calculate the force per unit length acting on wire Y.	
		force per unit length = N m ⁻¹	[2]
	L2	Magnetic force $F_{\text{on Y}} = B_{\text{at Y due to W}} I_{\text{in Y}} L_{\text{of Y}} \sin \theta$ where $\theta = 90^\circ$ since B and I are perpendicular. $\frac{F}{L} = BI$ $\frac{F}{L} = (1.5 \times 10^{-6})(1.0)$ $= 1.5 \times 10^{-6} \text{ N m}^{-1}$	M1 A1
	(iii)	Explain why the force that the two wires exert on each other is repulsive.	
			
			
			

				
				
				
				[3]
		L2	By Right Hand Grip Rule (or, using Fig. 2.1), the magnetic field produced by the current in wire W acts <u>perpendicular</u> to wire Y and <u>out of</u> the plane of paper. By Fleming's Left Hand Rule, the direction of the magnetic force <u>on Y by W</u> acts <u>to the right</u> / away from wire W. By Newton 3rd law of motion, the direction of the magnetic force <u>on W by Y</u> is <u>opposite to that on wire Y</u> / towards the left / away from wire Y. [OR, Apply Fleming's Left Hand Rule a second time to determine the force on W.] Therefore there is a repulsive force acting between the two wires.	M1 M1 M1 A0
		(iv)	Determine a possible position , other than at infinity, where the resultant magnetic flux density due to the magnetic fields of both wires is zero .	
			position:	[3]
		L2	First, consider directions of the two fields in different regions: - Net B cannot be zero between W and Y, since their fields point in the same direction. <u>On the left of both wires, and, on the right of both wires, their fields point in opposite directions.</u> Secondly, consider magnitude of B: - Since $B = \frac{\mu_0 I}{2\pi d} \Rightarrow B \propto \frac{I}{d}$. Hence Net B can only be zero <u>on the side further away from the larger current</u>. Thus, Net B is zero on the <u>right side of wire Y</u>. Let y be the distance to the <u>right of wire Y</u>.	M1 M1

		$ B_{\text{due to Y}} = B_{\text{due to W}} $ $\frac{\mu_0 (1.0)}{2\pi y} = \frac{\mu_0 (3.0)}{2\pi (40.0 \times 10^{-2} + y)}$ $\frac{(1.0)}{y} = \frac{(3.0)}{(40.0 \times 10^{-2} + y)}$ $40.0 \times 10^{-2} + y = 3.0y$ $2.0y = 40.0 \times 10^{-2}$ $y = 20.0 \times 10^{-2} \text{ m} = 20.0 \text{ cm}$ position: 20.0 cm to the right of wire Y. <i>(For marking: 'right' of wire Y can be stated either in the final answer space, or, working. Award A1 mark as long as this understanding that it must be right of Y be seen.)</i>	A1
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[Total: 12]

3	A uniform spherical star has a mass of $6.0 \times 10^{30} \text{ kg}$. The mass of the star may be assumed to be a point mass at the centre of the star. The star may be considered to be isolated in space.		
	(a)	Show that the gravitational field strength at a point $3.0 \times 10^9 \text{ m}$ from the centre of the spherical star is 44.5 N kg^{-1}.	
			[1]
	L1	$g = \frac{GM}{r^2} = \frac{(6.67 \times 10^{-11})(6.0 \times 10^{30})}{(3.0 \times 10^9)^2}$ $= 44.467 = 44.5 \text{ N kg}^{-1}$	M1 A0
	(b)	The radius of the star is $1.0 \times 10^9 \text{ m}$. On the axes of Fig. 3.1, sketch a graph to show the variation with distance x from the centre of the star of the gravitational field strength g of the star for values of x from $x = 1.0 \times 10^9 \text{ m}$ to $x = 4.0 \times 10^9 \text{ m}$.	